International Journal of Information Management 86 (2026) 102975

Contents lists available at ScienceDirect

International Journal of Information Management

journal homepage: www.elsevier.com/locate/ijinfomgt

Effect of corporate reputation and coopetitive branding strategy on digital products sales: Evidence from digital video game industry

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ARTICLE INFO

Keywords:
Digital video games
Corporate reputation
Coopetition
Coopetitive branding strategy
Digital product sales
Signaling theory

ABSTRACT

This research examines the impact of the corporate reputation of game developers and publishers and their coopetitive branding strategy on digital video game sales. The study utilizes empirical data from Steam, the world's largest digital video game distributor. Drawing on signaling theory, the study reveals the effect of two types of quality cues—corporate reputation and coopetitive branding strategy—on digital video game sales. The study finds the positive effect of both parties' corporate reputations on sales. Additionally, it finds the moderation effect of their reputation on the relationship between product popularity and sales, emphasizing the importance of corporate reputation in the digital video game market with enormous product diversity. Regarding a coopetitive branding strategy's effect, the findings disclose that only developer coopetitive branding—labeling multiple developer brands in a competitive relationship—effectively increases sales, while publisher coopetitive branding has no significant effect.

1. Introduction

Video gaming is one of the most popular leisure activities enjoyed by over 3 billion people (Adair, 2023), representing almost one-third of the world's population. The video game market generated $250 billion in revenue in 2023 and is expected to reach $363.2 billion by 2027 (Statista, 2023). Notably, the rise of digital video games—fee-based digital video games distributed via the Internet—has contributed significantly to this growth (Ubee, 2022). While physical copies of video games have seen declining revenue (from $16.1 billion in 2017 to $11.79 billion in 2023), digital video games have surged, reaching $222 billion during the same period (Rogers, 2023)—almost 19 times greater than physical video games. On online sales platforms, consumers can purchase various video games in a more convenient environment than in traditional shops. For example, the most dominant digital video game distributor, Steam, offers over 73 thousand games (Bray, 2024), and another large distributor, Green Man Gaming, has over 10 thousand games (GMG, 2024). Due to such overwhelming product diversity, however, consumers encounter information overload, which conversely inhibits effective purchase decisions (Chen et al., 2009) and encourages consumers to rely on quality cues of digital video games, signaling the potential quality of the products (Watson & Wu, 2022).

Based on signaling theory, prior research has disclosed the significant reliance of consumers on quality cues, such as eWOM (Gil & Warzynski, 2015), product popularity (Oh & Min, 2015), and user base size (i.e., the number of users) (Choi et al., 2018). Another frequently examined quality cue is the reputations of video game companies, although most of the literature reported its insignificant effect on sales (Choi et al., 2018; Gil & Warzynski, 2015; Nam & Kim, 2018; Yi et al., 2019). This does not correspond to the business strategy literature, claiming that consumers consider the reputations of product manufacturers and distributors as a signal of product quality (Dawar & Parker, 1994). These unexpected findings can be attributed to prior studies' problematic definitions of reputation developed based on a few companies (Choi et al., 2018; Cox, 2014; Nam & Kim, 2018), which can be improved with more granularly refined conceptualization. These prior studies also failed to distinguish developers and publishers, which are two critical parties in providing digital video game experience. Developers create video games, focusing on game story and algorithm design, graphic and sound design, and programming, while publishers conduct marketing, channeling, maintenance (e.g., updating and patching), and customer service, to deliver the video game experience to consumers. Although consumers should evaluate the reputation of both parties, given their unique roles in the upstream (i.e., development) and

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https://doi.org/10.1016/j.ijinfomgt.2025.102975
Received 1 April 2025; Received in revised form 11 August 2025; Accepted 30 August 2025
Available online 16 September 2025
0268-4012/Published by Elsevier Ltd.

downstream (i.e., publishing) of the digital video game lifecycle, prior studies considered only one of the two parties. This limitation could also offer little insights for practitioners who are looking for their potential partners. For instance, developers often deliberate whether to partner with large, reputable publishers for success, even though they should sacrifice their profits and vice versa (Logan, 2018). However, the findings in prior studies provide few practical implications due to the limitations. In addition, few studies examined a possible moderation effect of their reputation on the relationship between quality cues, such as product popularity, and digital video game sales. These research gaps offer opportunities not only to extend the knowledge of signaling theory but also to provide practical implications for boosting the sales of digital video games.

In the digital video game industry, companies often collaborate with their competitors to develop and publish games (Agate, 2023; Klimas & Czakon, 2018), a business strategy referred to as coopetition (Hong & Snell, 2015; Lioliou et al., 2025). According to Klimas and Czakon (2018), approximately 68 % of video game companies have adopted coopetition for diverse strategic advantages, including resource sharing, technology exchange, expertise collaboration, access to new markets and consumer segments, and cultivating innovation and creativity for video games (Agate, 2023). When releasing video games produced through coopetition, companies jointly expose their brands via the products and marketing campaigns, a business strategy known as 'coopetitive branding' (Sabri et al., 2021). Some prior research claimed potential benefits of coopetitive branding in sales due to the aggregated brand equity of alliances (Chiambaretto et al., 2016). However, little research empirically investigated corporate branding strategies, particularly coopetitive branding, as a quality cue for information asymmetry reduction and its subsequent impact on digital product sales, especially in the context of digital video games where a coopetitive branding strategy is widely adopted. As an attempt to address the gap in research and explore the components of signaling theory, this study examines the following questions: (1) *How do developer and publisher reputations impact the sales of digital video games?* (2) *How does their reputation impact the relationship between product popularity and sales?* (3) *How does the coopetitive branding strategy of developers and publishers affect sales?* This study aims to explore the research questions using an empirical market dataset. It includes nearly 312,000 observations for approximately 25, 200 digital video games, which is more sizable than those used in the prior studies examining the effect of video games' quality cues (Choi et al., 2018; Gil and Warzynski, 2015; Nam & Kim, 2018; Yi et al., 2019).

This research offers three key contributions to the literature of signaling theory and video games. First, this study expands the spectrum of quality cues by distinguishing the reputations of developers and publishers, which have seldom been considered in prior video game studies (Choi et al., 2018; Cox, 2014; Nam & Kim, 2018). This differentiation also extends the theoretical scope of signaling theory by illustrating that quality signals from distinct value chain entities have varying impacts on product sales, particularly within the IT industry. Second, it discovers the significant effect of both developers and publishers on digital video game sales and their positive moderating effect on other quality cue, which is product popularity in this study. Third, it suggests that coopetitive branding can be perceived as a quality cue from the consumers' perspective, which reduces information asymmetry (Gil & Warzynski, 2015; Yi et al., 2019), although the coopetitive branding between publishers may not be as critical as that between developers.

The remainder of this manuscript is structured as follows: the Literature review section discusses relevant literature and identifies research gaps, followed by the Hypothesis development section proposing hypotheses and a research model. The hypothesis testing process, including data collection and the research model, is discussed in the Research methodology section. The Data analysis section presents the results of hypothesis testing and robustness checks. Finally, the Discussion and contributions section explores primary findings and their implications, and the Limitations and future research section concludes the study with suggestions for other scholars interested in similar topics.

2. Literature review

This study investigates two key quality cues - the respective impact of the reputation of developers as well as publishers, and that of their coopetitive branding strategies on digital video game sales through the theoretical lens of signaling theory. Hence, the relevant literature to this objective is reviewed, such as the determinants of video game sales, corporate reputation's effect on sales, coopetition strategy's effect on business performance, and signaling theory.

2.1. Determinants of video game sales

In the extant literature, the determinants of video game sales can be classified into extrinsic and intrinsic factors. The former indicates the attributes that are not inherent to a product and, hence, easy to modify (Choi et al., 2018). The attributes, generally pertaining to product promotion strategies, can be further categorized according to their product lifecycle. Before the release of video games, their potential sales can be predicted by users' interest in internet search volume (Ruohonen & Hyrynsalmi, 2017) and the number of tweets (Malvankar et al., 2023). Video game release timing is another potential determinant, although prior studies have yielded mixed results. Some suggest a positive effect of avoiding competition with other video games (Gil & Warzynski, 2015), while others find no significant impact (Sacranie, 2010). After release, video game sales are highly determined by product marketing strategies, including post-release promotion (Gil & Warzynski, 2015), TV commercials (Yi et al., 2019), and discount pricing (Choi & Chen, 2019). Additionally, user-generated ratings (Binken & Stremersch, 2009; Gil & Warzynski, 2015; Sacranie, 2010), and the number of active users (Choi et al., 2018) can affect video game sales.

As an extrinsic quality cue, product popularity is one of the primary determinants of video game sales (Choi et al., 2018; Oh & Min, 2015). Because it enhances product visibility (Ifrach & Johari, 2014) and reduces consumers' uncertainty in the overcrowded market (Liu et al., 2014), consumers are more demanding of popular video games. In addition, higher popularity can attract more consumers at the early stage, which is particularly crucial in increasing sales and extending the life span of video games (Yi et al., 2019). However, few studies examined how the relationship between video game popularity and sales can vary based on the reputation of video game companies, including developers and publishers, although consumers perceive product quality differently according to corporate reputation (Eccles et al., 2007).

The intrinsic factors are concerned with inherent product attributes (Choi et al., 2018). For video games, they include video game platforms (e.g., Nintendo Wii, Xbox, PlayStation) and reputations of developers and publishers. Regarding the effect of platform variety, the extant literature commonly reports its positive relationship with video game sales (Babb & Terry, 2013; Corts & Lederman, 2009; Landsman & Stremersch, 2011) except for Sacranie (2010), which found no significant relationship. For instance, mobile app games on popular messenger platforms tend to achieve higher sales (Yi et al., 2019) due to their direct and indirect network externalities (Corts & Lederman, 2009), a phenomenon frequently observed in diverse IT markets.

Similarly, the reputation of product providers—developers and publishers in the context of video games—positively influences sales. Consumers often rely on manufacturer and retailer reputations as indicators of product quality (Dawar & Parker, 1994). In the literature on video games, Cox (2014) found that video games introduced by dominant companies, such as Activision, Nintendo, Rockstar, and Sony Entertainment, are likely to become blockbuster titles. However, the majority of literature reports no significant effect of the corporate reputation on video game sales (Choi et al., 2018; Gil & Warzynski, 2015; Nam & Kim, 2018; Yi et al., 2019). This is surprising because the reputation of manufacturers plays a critical role in determining product sales in a competitive market (Chun et al., 2005) like the video game industry, and reputable firms tend to have more resources for quality product (Readman & Grantham, 2006).

The mixed findings regarding the effect of the reputations of developers and publishers on video game sales could be due to several reasons, offering research opportunities for the present study. First, researchers focused on a few well-known companies when examining the impact of company reputation on video game sales. For instance, Choi et al. (2018) defined 19 digital video game companies listed on metacritic.com as the major firms and investigated whether digital video games from these 19 companies generate higher sales. Similarly, Nam and Kim (2018) selected only three firms without specific reasons (e.g., revenue, firm size) and tested whether digital video games from these three companies have higher sales. Additionally, they failed to consider both developers and publishers, although the reputation of each party could have a distinct impact on digital video game sales. For example, Gil and Warzynski (2015) considered only the developers' reputation, while Nam and Kim (2018) focused solely on the publishers. Due to these limitations, it is still unclear how the corporate reputation of each party affects digital video game sales.

2.2. Effect of corporate reputation on product sales

Brand reputation reflects how outsiders perceive a company based on its past actions, shaping their future expectations (Chierici et al., 2019; Sun et al., 2024). Brand reputation influences attitudes toward brands and impacts product sales, enabling consumers to recognize quality products (Azzari & Pelissari, 2021; Sun et al., 2024). While earlier research highlights various facets of brand reputation, it is commonly understood to include three core dimensions: familiarity with the brand, beliefs about the brand's future actions, and impressions about the brand's favorability (Lange et al., 2011). Brand familiarity encompasses awareness of brand name, identity, and core attributes (Barnett et al., 2006; Liu & Xiong, 2023). Beliefs about the brand involve perceived quality and managerial competence in upholding brand values (Hur et al., 2014; Jensen & Roy, 2008). Brand favorability, on the other hand, reflects affective evaluation of consumers and appeals to them against competitors (Doh et al., 2010; Rhee & Valdez, 2009). These three dimensions represent a valuable asset, offering a range of advantages such as attracting quality talent, enabling strategic agility, and enhancing market performance (Lange et al., 2011).

In business literature, brand reputation can be categorized into corporate and product reputation. For example, the iPhones are linked to Apple Inc.'s corporate reputation and the product reputation of its previous models. Product reputation directly affects purchase intent (Azzari & Pelissari, 2021), product sales, and market share (Huang & Sarigöllü, 2012). However, the impact of corporate reputation on business performance remains inconclusive. Some studies report its significant effect on purchase intent (Cretu & Brodie, 2007) and product sales (Huang & Sarigöllü, 2012), while others argue that it does not directly affect purchase intention (Lee & Lee, 2018). These studies have explored the impact of corporate reputation in different markets, such as hotels (Anggani & Suherlan, 2020), restaurants (Han et al., 2011a), fashion products (Jung & Seock, 2016; Lee & Lee, 2018), and fast-moving consumer goods (Cretu & Brodie, 2007; Huang & Sarigöllü, 2012). These inconsistent findings imply that the effect may differ by market domains. Therefore, they may not be applicable to IT domains, which necessitates further investigation into the effect within the digital video game market.

2.3. Effect of coopetition strategy on business performance

Among diverse types of business collaboration strategies, coopetition—indicating collaboration with competitors—has been growing across different business domains (Hong & Snell, 2015), particularly the IT industry (Zapadka et al., 2022). Accordingly, the effect of coopetition has received great attention in the academic literature. The studies can be divided by different business performance metrics, such as organizational innovation (Chai et al., 2020), product innovation (Bouncken et al., 2018), a firm's financial performance (Crick et al., 2024), and product sales (Bahar et al., 2022). Although the studies generally agree with the positive effect of coopetition, they suggest that the effect depends on conflict and balance between firms (Chai et al., 2020; Park et al., 2014), market environment (Ritala, 2012), project uncertainties and complexities (Un & Asakawa, 2015), and reliance on partners (Zulu-Chisanga et al., 2025).

Depending on when coopetition arises during the product lifecycle, it can manifest as either upstream or downstream coopetition (Rusko, 2015). Upstream coopetition primarily occurs during research and development (R&D) or when acquiring and processing raw materials. In contrast, downstream coopetition focuses on developing new offerings and performing promotional activities in the market. Coopetition can be further categorized based on whether the multiple competitors involved in a project expose their brands (e.g., trademarks) through the product or service, an outcome of their collaboration (Sabri et al., 2021). When companies jointly label their brands, it is referred to as coopetitive branding, indicating a business strategy in which two or more independent companies in a potentially competitive relationship present themselves on a product or service. According to signaling theory, coopetitive branding can enhance signals that increase product (or service) awareness and trust due to a trust transference effect. When multiple companies align for a product, they can increase their product awareness, positive perceptions, and trust in their products, because consumers would more readily trust the product associated with multiple companies that they have confidence in (Mavlanova et al., 2016). However, the effect of coopetitive branding has been seldom discussed in the extant literature.

Due to the increasing importance of coopetition, recent IS studies have investigated its impact on business performance, leaving several notable research gaps. First, their findings on the coopetition's effect are contradictory. Some studies claim its positive impact, arguing that coopetition enhances open innovation performance and capacity to innovate (Hameed & Naveed, 2019) because this collaboration strategy allows firms to benefit from more technical diversity and a wider range of product categories (Quintana-Garcia & Benavides-Velasco, 2004) and to compensate for each other's weaknesses (Lioliou et al., 2025). In contrast, other studies suggest a negative effect, as coopetition may hinder radical technology innovation (Ritala & Sainio, 2014), the development of novel product functions (Nieto & Santamaría, 2007), and overall product innovation among manufacturing firms (Un & Asakawa, 2015; Un et al., 2010). Consequently, its impact on firm performance, including product sales, remains an ongoing debate in the literature (Gernsheimer et al., 2021). Second, prior research has focused on upstream coopetition, except for Robert et al. (2018), exploring how coopetition between real estate agents affects the sales of their listed properties, which occurs in the downstream phase. Lastly, little research empirically examined the impact of coopetitive branding on product sales, although the effect of this strategy can vary depending on where coopetition occurs in the product lifecycle. For instance, the impact of coopetitive branding in upstream coopetition (e.g., video game development) can be more or less substantial than that in downstream coopetition (e.g., video game publishing). However, although widely embraced in the IT industry, research exploring these distinctions remains scarce, especially in the digital video game sector, where coopetition strategies are commonly employed (Klimas & Czakon, 2018).

2.4. Signaling theory

Signaling theory is widely used to explain the behavior of two parties interacting with a task with different information (i.e., information asymmetries) (Nishant et al., 2023). In business transactions, signaling theory posits that buyers, who typically have less information than sellers, infer product quality from various cues. This phenomenon is more pronounced in e-commerce environments where consumers cannot physically examine products but instead rely on the information provided through e-commerce platforms (Mavlanova et al., 2012).

Due to the importance of quality cues in e-commerce, prior studies have investigated the impact of various signals on e-commerce consumers, with a particular focus on the distinctive features of e-commerce environments. These cues encompass perceived website quality (Gregg & Walczak, 2008; Wells et al., 2011), online product reviews (Chen et al., 2020; Cheung et al., 2014), online seller reputation (Al-Adwan & Yaseen, 2023; Ou & Chan, 2014; Su, 2007), quality of real-time communication with sellers (Dong et al., 2024; Lu & Chen, 2021), product recommendation (Chen et al., 2019; Li et al., 2024), and trust badges for online sellers (Aiken & Boush, 2006; Lladós-Masllorens et al., 2020). As these signals are more positive, consumers will likely engage in transactions with sellers on e-commerce platforms.

In video game literature, recent studies have applied signaling theory to explore the dynamics of corporate reputation and product popularity. Rodrigues et al. (2024) used the theory to demonstrate how in-game advertising of luxury brands influences their brand reputations, revealing that the signals emitted through in-game advertising directly or indirectly enhance brand coolness, awareness, perceived quality, and loyalty. Similarly, Tang and He (2022) found that publisher credibility, game genre, review rating and quantity, recommendation frequency, and game ranking (i.e., popularity) determine mobile game downloads, as the attributes play as quality cues. A more relevant study to this present research is Choi et al. (2018). They used the theory to examine the effect of corporate reputation and product popularity on digital video game sales. However, they treated the reputation as a single quality cue without distinguishing between developers and publishers, which could be a potential reason why they found no significant relationship between corporate reputation and sales.

2.5. Conclusion of literature review

As illustrated, although extant literature has discussed the determinants of video game sales and the effect of corporate reputation on product sales, few have considered the respective impact of the corporate reputation of developers and publishers on digital video game sales. In terms of signaling theory, existing research has overlooked the specific impact of developers' and publishers' reputations on IT product sales and their potential moderating effect on other quality cues, even though the two parties play different roles in the product lifecycle. Additionally, little prior research has considered coopetitive branding in the development and publishing stages as an essential quality cue that provides relevant information for consumers who cannot fully understand video game quality, despite its growing popularity in the industry (Klimas & Czakon, 2018).

3. Hypothesis development

3.1. Effect of corporate reputation on digital video game sales

A corporate reputation refers to a conceptual representation of a business organization's past actions and expectations for its future actions compared with those of its competitors (Veh et al., 2019). Although its indirect relationship between reputation and product sales through product satisfaction has been established (Susanti & Samudro, 2022), its direct impact on sales remains a topic of debate in existing literature: some studies suggest positive (Cretu & Brodie, 2007), while others find it insignificant (Huang & Sarigöllü, 2012).

Despite these mixed findings, this present study predicts that the corporate reputation of developers and publishers plays a significant role in determining digital video game sales as a cue signaling their potential quality. Drawing from signaling theory (Shapiro, 1983), consumers often place greater reliance on quality cues such as corporate reputation when making purchase decisions, especially in an overcrowded market like the digital video game industry. For example, Steam, the most prominent digital video game distributor globally, currently offers over 73 thousand video games (Bray, 2024), with more than 14 thousand new games introduced in 2023 (Obedkov, 2023). In such a competitive market, consumers should pay significant attention to both developers and publishers of digital video games, as each party determines game quality during development (e.g., graphic and character design, game story development, etc.) and service (e.g., updating, patching, and customer service).

In addition, the impact of corporate reputation on sales is more evident for experience products than search products. As a typical experience product, video game quality (Zhu & Zhang, 2006) cannot be fully assessed before actual consumption (Grunwald & Hempelmann, 2010). Hence, consumers should give due consideration to the reputation of developers and publishers, as it acts as an assurance of video game quality. This discussion leads to the following hypotheses:

H1a: Developer reputation has a positive relationship with digital video game sales.

H1b: Publisher reputation has a positive relationship with digital video game sales.

3.2. Moderation of corporate reputation on the relationship between product popularity and sales

It is generally accepted that a product's perceived quality, as an essential quality cue, significantly influences both purchase intention and sales (Tsiotsou, 2006). As a critical aspect of perceived quality, product popularity positively impacts perceived quality and purchase intention (Jeong & Kwon, 2012). In e-commerce, product popularity serves as a primary determinant of product sales (Chen, 2008; Duan et al., 2009), including digital video game sales (Choi et al., 2018). Beyond the direct effects (Hypotheses 1a and 1b), the reputation of developers and publishers is expected to strengthen the relationship between product popularity and digital video game sales.

According to brand strategy literature grounded in signaling theory, the influence of perceived product quality on sales can be even more prominent when the product possesses additional advantages across other dimensions of brand quality, which act as quality signals (Cowan & Guzman, 2020; Hirvonen et al., 2013). For example, perceived brand quality reinforces the association between brand heritage and consumers' willingness to pay premium prices (Pecot et al., 2018). Similarly, when customers perceive a high level of corporate social responsibility, which positively affects corporate reputation, it enhances the impact of perceived service quality on customer satisfaction. The satisfaction, in turn, contributes to improved business performance (Yuen et al., 2016). This implies that when consumers contemplate purchasing a popular digital video game, they are more likely to choose it if it is developed by reputable companies, all else being equal.

Additionally, digital video games, as software, require consistent updates and maintenance for quality assurance (Erdil et al., 2003). When considering buying a popular digital video game, consumers are more likely to purchase it if offered by well-established publishers with the resources to deliver reliable updates and patches. This discussion predicts the following hypotheses:

H2a: The positive relationship between product popularity and sales is larger for digital video games of reputable developers.

H2b: The positive relationship between product popularity and sales is larger for digital video games of reputable publishers.

3.3. Effect of coopetitive branding strategy on digital video game sales

As discussed, coopetition occurs in both upstream and downstream stages of the video game lifecycle. Game development aligns with the upstream phase, whereas publishing and maintenance fall downstream

(Rusko, 2015). Therefore, coopetition among developers leads to coopetitive branding on the upstream, while that among publishers leads to coopetitive branding on the downstream. As a quality cue, a coopetitive branding strategy can widen market segments, as well as enhance product awareness and credibility (Chiambaretto et al., 2016).

Examining the impact of coopetitive branding on digital video game sales, this study predicts a positive effect of the business strategy for both developers (i.e., upstream) and publishers (i.e., downstream). Signaling theory postulates that the presence of corporate alliances enhances product awareness, trust, and perceived quality (Mishra et al., 2017) through a trust transference (Mavlanova et al., 2016). When the signals of multiple credible firms are incorporated into a product, consumers tend to perceive the product as having higher quality (Fang et al., 2013). Therefore, consumers would perceive digital video games with multiple developers or publishers as having higher quality due to the additional credibility. In addition, co-branded games can tap into broader customer segments across multiple companies (Mróz-Gorgoń, 2016), resulting in higher sales.

In terms of business strategy literature, companies engaged in coopetition can leverage functional diversity by learning each other's technical expertise (Van Beers & Zand, 2014) or cultivate productive inter-firm interdependencies by sharing similar tasks (Wiener & Saunders, 2014). In game development, coopetition allows them to collectively enhance game design, sound engineering, and quality assurance by utilizing shared expertise. In the publishing stage, it enables the distribution of video games to more geographical areas through diverse video gaming platforms (e.g., mobile, console, PC). For instance, Ubisoft partnered with Tencent, which is one of its major competitors, to publish video games in China (Calvin, 2018), enabling them to penetrate the largest video game market globally. Therefore, this collaborative effort not only ensures video game quality but also positively impacts sales, proposing the following hypotheses:

H3a: Developer coopetitive branding has a positive relationship with digital video game sales.

H3b: Publisher coopetitive branding has a positive relationship with digital video game sales.

Fig. 1 below summarizes the hypotheses proposed:

4. Research methodology

4.1. Data collection

This study collected data from three sources to test the proposed hypotheses: steamspy.com, store.steampowered.com, and vginsights.com. Steamspy.com is a third-party website that provides sales volume data for digital video games on store.steampowered.com, the largest digital video game platform globally (Koutsou-Wehling, 2024). Steamspy's sales data is highly accurate, with an error rate of less than 0.33 % compared to actual data (Gilbert, 2015), although access to this data is limited to paid members. Store.steampowered.com serves as Steam's sales platform. It offers diverse digital video game data, including genre, release date, price, developers, and publishers. Lastly, vginsights.com is a video game market research platform offering data related to digital video games, developers, and publishers. Data collection occurred daily from February 21, 2023, to September 17, 2023, for 208 days. Because steamspy.com, which is the data source of daily sales, provides sales data of video games with at least 2000 daily sales, video games with fewer than 2000 daily sales were excluded from the dataset. Additionally, approximately 1100 video games were excluded due to missing developer or publisher information, such as accumulated total revenue and median revenue. These are primarily small or indie companies. As a result, the dataset for this study comprises 311,650 observations for 25,190 digital video games developed or published by 974 companies.

4.2. Operationalization of constructs

In this study, the dependent variable—*Daily Sales*—is operationalized as the daily sales volume, representing the number of daily downloads for each digital video game. Concerning the two primary independent variables—*Developer Reputation* and *Publisher Reputation*—this study operationalizes them based on the accumulated total revenue of a video game company. Financial performance has been widely used as a critical measure of corporate reputation (Chun, 2005; Lewis, 2001), particularly in relation to beliefs about the brand and brand favorability (Lange et al., 2011).

The top ten percent performing firms in terms of their revenue, often regarded as dominant firms in a market domain (Mirocha et al., 2013; Rarick & Nickerson, 2006), are defined as reputable companies in the market. According to their roles in the video game lifecycle—either developing or publishing—if a company belongs to the top ten percent of companies in terms of their total revenue, it is coded as 1 (otherwise, 0) in *Developer Reputation* and *Publisher Reputation*. *Developer Coopetitive Branding* and *Publisher Coopetitive Branding* are operationalized based on the definition of coopetitive branding, *"a voluntary strategy that consists of combining and presenting jointly two or more independent brands from competing firms on a product or service"* (Chiambaretto et al., 2016). When more than one company in a potentially competitive relationship has been involved in developing (or publishing) a game, and their brands are labeled on the video game, they are coded as 1. Lastly, *Product Popularity* is estimated by a daily ranking of digital video games in the market (Zhu & Zhang, 2010), a reverse measure of popularity in that smaller numbers in the ranking list indicate higher popularity.

In addition to the main variables, this study incorporates several control variables known to affect digital video game sales to address potential endogeneity, including *User Base, Game Age, Price, Discount,* and *Update. User Base* represents the total number of game users (i.e.,

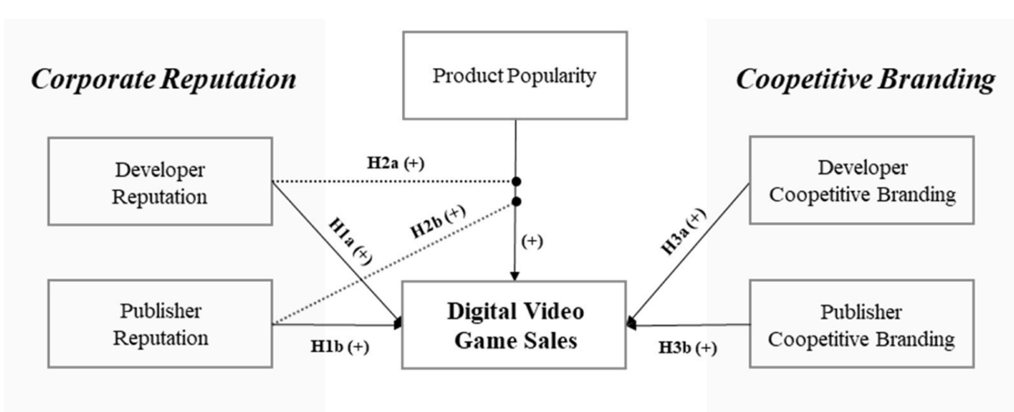

Fig. 1. Conceptual model of hypotheses.

network externalities), which is estimated by the accumulated unit sales of a video game. *Game Age* indicates how long a game has been available in the market, which is measured based on the first release date of a video game. *Price* refers to the actual price in USD, while Discount is a dummy variable indicating whether a video game was offered at a discounted price at the point of sale. *Update* refers to whether a video game was updated on a particular day. If updated, it is coded as 1 and otherwise 0. Given the impact of game genres on sales (Tang & He, 2022), the following dummy variables are included: *Casual, Action, Adventure, Early Access, Ex Early Access, MMO, RPG, Simulation, Sports,* and *Strategy*. These are determined based on the genre information reported by video game companies.

4.3. Research model

The econometric model presented below aims to test the proposed hypotheses, incorporating the specified variables and interaction terms.

$$\begin{aligned}
&Ln(Daily\ Sales_{i,t}) = \alpha_0 + \alpha_1 Developer\ Reputation_{i,t-1,} + \alpha_2 Publisher\ Reputation_{i,t-1} + \\
&\alpha_3 Developer\ Coopetitive\ Branding_i + \alpha_3 Publisher\ Coopetitive\ Branding_i + \\
&\alpha_4 Product\ Popularity_{i,t-1} + \alpha_5 Developer\ Reputation_{t-1} {}^{*}\ ProductPopularity_{i,t-1} + \\
&\alpha_6 Publisher\ Reputation_{t-1*} {}^{*} Product\ Popularity_{i,t-1} + \alpha_7 User\ Base_{i,t-1} + \alpha_8 Game\ Age_{i,t-1} + \\
&\alpha_9 Price_{i,t-1} + \alpha_{10} Discount_{i,t-1} + \alpha_{11} Update_{i,t-1} + \alpha_{12} Casual_i + \alpha_{13} Action_i + \alpha_{14} Advenutre_i + \\
&\alpha_{15} Early\ Access_i + \alpha_{16} Ex\ Early\ Access_i + \alpha_{17} MMO_i + \alpha_{18} RPG_i + \alpha_{19} Simulation_i + \\
&\alpha_{20} Sports_i + \alpha_{21} Strategy_i + \varepsilon_{i,t}
\end{aligned}$$

where *i* indicates video game and *t* does the date

The model includes two interaction terms, *Developer Reputation*-Product Popularity* and *Publisher Reputation*Product Popularity,* to examine Hypothesis 2a and Hypothesis 2b for the interaction effect of the reputation of developers and publishers on the relationship between product popularity and sales. Given the skewed distribution of *Daily Sales*, natural log transformation is applied to the variable, converting it to Ln *(*Daily Sales).

Prior studies have emphasized that the impact of IT product attributes displayed on e-commerce websites should be assessed over an extended time frame because the effect may not immediately occur (Choi et al., 2018; Lee & Raghu, 2014; Liu et al., 2014; Xie et al., 2016). Consequently, the studies employed the explanatory variables at t-1 (day-1) while using the target variable at t, which is Ln *(*Daily Sales$_{i,t}$*)*, in this study. However, the lagged effect is not considered for the variables related to video game genres, as they remain constant over time.

5. Data analysis

5.1. Descriptive statistics

Table 1 summarizes the descriptive statistics of the variables in the research model. The statistics reveal that 6.4 % of observations in the data are digital video games developed by reputable developers (i.e., top 10 % developers), while 17.6 % of the games are published by such publishers (i.e., top 10 % publishers). Regarding coopetitive branding, 10.5 % of the games have more than one developer, while 8.0 % are published by multiple companies. These findings indicate the presence of a coopetitive branding strategy in the digital video game industry.

The correlation matrix in Table 2 illustrates no significant multicollinearity among variables. Further, the Variance Inflation Factor (VIF) test is administered to check possible multicollinearity (Mansfield & Helms, 1982). Consistent with observation in the correlation matrix, the most considerable VIF value is 4.32 for *Publisher Reputation*, which is notably below the threshold of 10 (Naser et al., 2013), and the average VIF of all variables is 1.66, confirming no multicollinearity.

5.2. Hypothesis test results

Since the dataset for this research is in the form of panel data, tracking changes in digital video games over time, it is prone to violating the assumptions of Ordinary Least Squares (OLS). To address this potential concern, a Breusch-Pagan test is used to check heteroscedasticity. Not surprisingly, the results confirm the presence of heteroscedasticity in the dataset ($p < 0.01$), requiring a more robust estimator beyond the OLS regressor. Therefore, this study adopts a random effects model with robust standard errors (Greene, 2003), an estimation previously employed by Information Systems studies that used similar panel data (Choi et al., 2018; Liu et al., 2014; Mallapragada et al., 2016), as well as allowing time-invariant variables in the model (Kathuria et al., 2023). Table 3 summarizes the analysis results obtained from the random effects model.

The coefficient of *Developer Reputation* $_{i,t\text{-}1}$ (α_1) is positive (0.2487) and significant ($p = 0.03$). This result aligns with Hypothesis 1a, which predicts a positive relationship between developer reputation and digital video game sales. Hypothesis 1b is also supported because the coefficient of *Publisher Reputation* $_{i,t\text{-}1}$ (α_2) is positive (0.2489) and significant ($p < 0.01$).

Next, the interaction of the corporate reputation of developers and publishers on the relationship between product popularity and sales is tested with *Developer Reputation* $_{i,t\text{-}1}$**Product Popularity* $_{i,t\text{-}1}$ and *Publisher*

Table 1
Descriptive statistics.

Variable	Mean	Standard Deviation	Minimum	Maximum
Daily Sales	24394.920	127732.100	2000	12700000
Product Popularity	775.228	482.120	1	2955
User Base	887129.600	3435426.000	0	81000000
Game Age	2634.632	2129.517	0	14690
Price	13.880	13.677	1	200
Frequency (n = 311650)				
Developer Reputation	20059 (6.4 %)			
Publisher Reputation	55003 (17.6 %)			
Developer Coopetitive Branding	32825 (10.5 %)			
Publisher Coopetitive Branding	24981 (8.0 %)			
Discount	72240 (23.2 %)			
Update	1668 (0.5 %)			
Casual	3772 (1.2 %)			
Utility	1663 (0.5 %)			
Action	155087 (49.8 %)			
Adventure	128246 (41.2 %)			
Early Access	20027 (6.4 %)			
Ex Early Access	21128 (6.8 %)			
MMO	6720 (2.2 %)			
RPG	65307 (21.0 %)			
Simulation	65261 (20.9 %)			
Sports	12590 (4.0 %)			
Strategy	75975 (24.4 %)			

Table 2
Correlation matrix.

VAR	DS	UB	PP	GA	PE	DR	PR	DC	PC	DT	CA	UT	AT	AD	EA	EE	RP	SM	SP	ST	UP
DS	1.00																				
UB	0.18	1.00																			
PP	−0.17	−0.35	1.00																		
GA	0.06	0.01	−0.09	1.00																	
PE	0.05	0.21	−0.39	−0.21	1.00																
DR	0.10	0.27	−0.29	0.09	0.33	1.00															
PR	0.09	0.27	−0.42	0.10	0.48	0.50	1.00														
DC	0.02	0.02	−0.08	0.04	0.10	0.08	0.12	1.00													
PC	0.01	0.03	−0.06	0.10	0.08	0.07	0.00	0.28	1.00												
DT	−0.01	0.01	−0.07	−0.06	0.05	0.03	0.05	0.01	−0.01	1.00											
CA	−0.02	−0.03	0.07	0.04	−0.06	−0.02	−0.04	−0.03	−0.01	−0.01	1.00										
UT	−0.01	−0.01	0.00	0.00	0.03	−0.02	−0.03	−0.01	−0.02	−0.02	−0.01	1.00									
AT	0.05	0.10	−0.14	−0.06	0.11	0.08	0.14	0.02	−0.02	0.02	−0.10	−0.07	1.00								
AD	0.00	0.01	0.04	−0.08	0.02	−0.10	−0.06	−0.01	−0.01	0.02	−0.07	−0.06	0.06	1.00							
EA	−0.01	0.02	0.02	−0.16	0.02	−0.07	−0.10	−0.03	−0.01	−0.03	−0.03	−0.02	0.06	0.02	1.00						
EE	0.03	0.09	−0.16	−0.12	0.08	−0.02	−0.03	0.01	−0.01	0.02	−0.03	−0.02	0.05	−0.01	−0.01	1.00					
RP	0.04	0.17	−0.07	−0.07	0.06	0.01	0.02	0.03	0.00	−0.01	−0.02	−0.01	0.09	0.03	0.13	0.07	1.00				
SM	0.02	0.08	−0.07	−0.09	0.10	0.04	0.04	−0.01	0.01	0.01	−0.05	−0.04	−0.01	0.15	0.06	0.08	0.06	1.00			
SP	0.00	0.01	0.00	−0.04	0.04	−0.08	−0.09	0.00	0.00	−0.01	−0.05	−0.04	−0.15	−0.12	0.11	0.11	0.08	−0.04	1.00		
ST	0.00	0.00	0.04	−0.04	0.03	−0.03	−0.03	−0.01	−0.01	0.00	−0.01	−0.02	−0.02	−0.08	0.02	0.04	0.05	−0.06	0.21	1.00	
UP	−0.01	−0.03	0.00	0.04	0.02	0.00	−0.01	0.04	0.03	−0.02	−0.06	−0.04	−0.22	−0.19	0.02	0.06	0.05	0.06	0.22	−0.01	1.00

DS: Daily Sales, **UB:** User Base, **PP:** Product Popularity, **GA:** Game Age, **PE:** Price, **DR:** Developer Reputation, **PR:** Publisher Reputation, **DC:** Developer Coopetitive Branding, **PC:** Publisher Coopetitive Branding, **DT:** Discount, **CA:** Casual, **UT:** Utility, **AT:** Action, **AD:** Adventure, **EA:** Early Access, **EE:** Ex Early Access, **RP:** RPG, **SM:** Simulation, **SP:** Sports, **Strategy:** ST, **UP:** Update

Table 3
Analysis results.

Dependent: Ln (Daily Sales $_{i,t}$)	Coefficient	Standard Error	p-value
Constant	8.9386	0.0223	0.01 >
Developer Reputation $_{i,t-1}$	0.2487	0.1222	0.03
Publisher Reputation $_{i,t-1}$	0.2489	0.0646	0.01 >
Product Popularity $_{i,t-1}$	−0.0006	0.0001	0.01 >
Developer Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0004	0.0001	0.01 >
Publisher Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0466	0.0219	0.03
Publisher Coopetitive Branding $_i$	−0.0189	0.0332	0.57
Controls			
User Base $_{i,t-1}$	0.0001	0.0001	0.01 >
Game Age $_{i,t-1}$	0.0001	0.0001	0.01 >
Price $_{i,t-1}$	0.0036	0.0007	0.01 >
Discount $_{i,t-1}$	−0.0257	0.0074	0.01 >
Update $_{i,t-1}$	0.1909	0.0339	0.01 >
Casual $_i$	−0.2310	0.0327	0.01 >
Utility $_i$	−0.2392	0.0684	0.01 >
Action $_i$	0.0731	0.0119	0.01 >
Adventure $_i$	0.0640	0.0118	0.01 >
Early Access $_i$	−0.0662	0.0197	0.01 >
Ex Early Access $_i$	0.3271	0.0375	0.01 >
MMO $_i$	0.1602	0.0653	0.01 >
RPG $_i$	0.1194	0.0152	0.01 >
Simulation $_i$	0.0392	0.0152	0.01 >
Sports $_i$	−0.0600	0.0277	0.03
Strategy $_i$	0.0350	0.0141	0.01 >

Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$. Regarding the main effect of product popularity, the coefficient of *Product Popularity* $_{i,t-1}$ (α_4) is negative (-0.0006) and statistically significant ($p < 0.01$). This indicates that digital video games at higher ranks, coded with smaller numbers, tend to have higher sales. The coefficient of *Developer Reputation* $_{i,t-1}$**Product Popularity* $_{i,t-1}$ (α_5) is negative (-0.0004) and significant ($p < 0.01$), supporting Hypothesis 2a. This suggests that the positive effect of product popularity on sales is greater for the digital video games developed by reputable companies in terms of their total revenue.

Hypothesis 2b is also supported, given the negative (-0.0003) and significant ($p < 0.01$) coefficient of *Publisher Reputation* $_{i,t-1}$**Product Popularity* $_{i,t-1}$ (α_6). This attests to the positive interaction of publisher reputation on the relationship between product popularity and digital video game sales. The two moderation effects are also evident in the predictive margin plots in Fig. 2. Digital video game sales of the top 10 % of developers and publishers, defined as reputable companies in this study, consistently outperform those of other developers and publishers across different levels of product popularity.

Hypothesis 3a is tested with *Developer Coopetitive Branding* $_i$ in the research model. Its coefficient is positive (0.0502) and significant ($p = 0.03$), supporting Hypothesis 3a. This illustrates that video games created collaboratively by multiple developers, featuring their distinct brands, such as logos and trademarks, achieve greater sales. However, Hypothesis 3b, postulating the positive relationship between publisher coopetitive branding and sales, is not supported because *Publisher Coopetitive Branding* $_i$ is not statistically significant ($p = 0.57$).

Fig. 3 below summarizes the hypothesis results discussed.

5.3. Robustness check

Several robustness checks are performed to validate the findings of the hypothesis test. First, the Wooldridge test is performed to check for serial correlation in the dataset ($p < 0.01$). It indicates the presence of AR1 autocorrelation, and hence, a random effects estimation with AR1 errors is deployed to address the autocorrelation (Bardhan et al., 2013; Han et al., 2011b). As illustrated in Table 4, the hypothesis results remain unchanged from those of the main analysis (*c.f.,* controls are

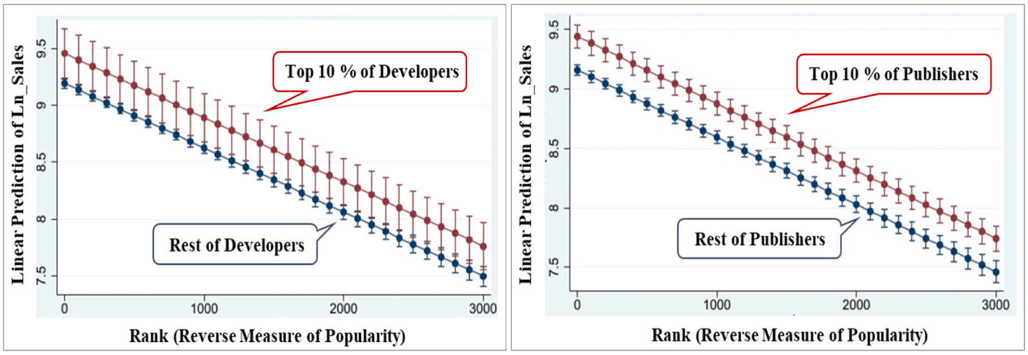

Fig. 2. Predictive margins plots: moderation of developer and publisher reputation.

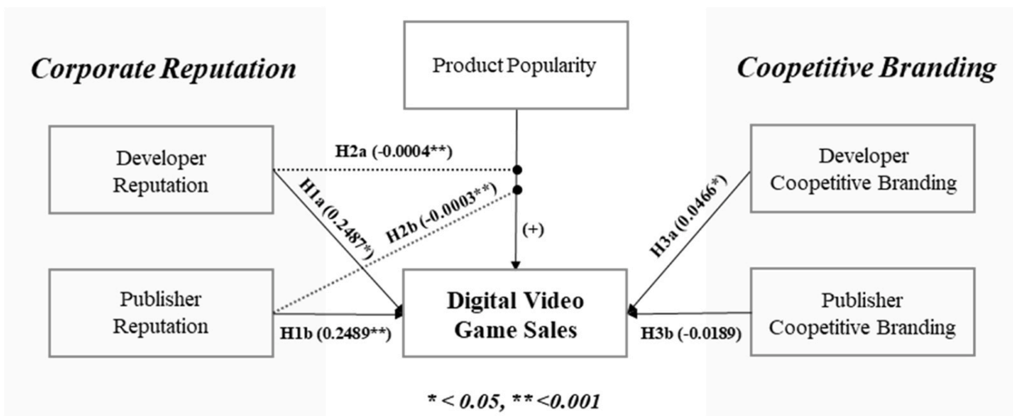

Fig. 3. Hypothesis test results.

Table 4
Analysis results of random effects with AR1 errors.

Dependent: Ln (Daily Sales $_{i,t}$)	**Coefficient**	**Standard Error**	**p-value**
Constant	8.7831	0.0198	0.01 >
Developer Reputation $_{i,t-1}$	0.2952	0.0521	0.01 >
Publisher Reputation $_{i,t-1}$	0.2907	0.0324	0.01 >
Product Popularity $_{i,t-1}$	−0.0004	0.0001	0.01 >
Developer Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0004	0.0001	0.01 >
Publisher Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0446	0.0201	0.03
Publisher Coopetitive Branding $_i$	−0.0275	0.0231	0.23

Table 5
Analysis results of random effects with bootstrapping (200 Times).

Dependent: Ln (Daily Sales $_{i,t}$)	**Coefficient**	**Standard Error**	**p-value**
Constant	8.9386	0.0238	0.01 >
Developer Reputation $_{i,t-1}$	0.2486	0.1208	0.04
Publisher Reputation $_{i,t-1}$	0.2488	0.0711	0.01 >
Product Popularity $_{i,t-1}$	−0.0005	0.0001	0.01 >
Developer Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >
Publisher Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0465	0.0211	0.02
Publisher Coopetitive Branding $_i$	−0.0189	0.0358	0.59

Table 6
Analysis results of random effects with jackknife estimator.

Dependent: Ln (Daily Sales $_{i,t}$)	**Coefficient**	**Standard Error**	**p-value**
Constant	8.9386	0.0272	0.01 >
Developer Reputation $_{i,t-1}$	0.2657	0.1361	0.05
Publisher Reputation $_{i,t-1}$	0.2824	0.0772	0.01 >
Product Popularity $_{i,t-1}$	−0.0005	0.0001	0.01 >
Developer Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0004	0.0001	0.01 >
Publisher Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0486	0.0221	0.02
Publisher Coopetitive Branding $_i$	−0.0153	0.0335	0.64

Table 7
Analysis results of random effects with the Hausman-Taylor estimator.

Dependent: Ln (Daily Sales $_{i,t}$)	**Coefficient**	**Standard Error**	**p-value**
Constant	7.9268	0.1436	0.01 >
Developer Reputation $_{i,t-1}$	1.2818	0.3578	0.01 >
Publisher Reputation $_{i,t-1}$	0.9190	0.2005	0.01 >
Product Popularity $_{i,t-1}$	−0.0001	0.0001	0.01 >
Developer Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0005	0.0001	0.01 >
Publisher Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0004	0.0001	0.01 >
Developer Coopetitive Branding $_i$	4.6525	0.7044	0.01 >
Publisher Coopetitive Branding $_i$	−3.7613	0.2476	0.01 >

omitted in the following tables for brevity).

Further, bootstrapping and jackknifing are applied to the estimator to further account for the heteroskedasticity detected and serial correlation within the unbalanced panel data (Thombs & Schucany, 1990). As shown in Table 5 and Table 6, they show consistent results with the main analysis.

Second, because predictors in the model can be correlated with the unobserved individual-level effects, the Hausman-Taylor instrumental variable estimator is employed to address the potential endogeneity (Hausman & Taylor, 1981), which allows the model to specify and estimate endogenous variables. In this estimation, the predictors pertaining to proposed hypotheses, such as *Developer Reputation, Publisher Reputation, Developer Coopetitive Branding,* and *Publisher Coopetitive Branding,* are specified as endogenous variables. For instance, developer/publisher reputation, which is operationalized with company revenue, can be influenced by business strategies, while a coopetitive branding strategy can be affected by corporate culture (e.g., openness to collaboration with competitors). Table 7 shows mostly consistent results as in the main analysis. A noticeable difference is a negative coefficient (-3.7613) of *Publisher Coopetitive Branding* ($p < 0.01$), which was insignificant in the previous analyses. However, this is consistent in that the result does not correspond to the prediction of Hypothesis 3b.

Third, one of the critical aspects of the main analysis design is the definition of *Developer Reputation* and *Publisher Reputation,* as it substantially impacts the test results of the hypotheses tied to this definition, including Hypotheses 1a, 1b, 2a, and 2b. The main analysis defined *Developer Reputation* and *Publisher Reputation* as the top 10 % of video game companies based on their total revenue. To check the sensitivity to this definition, a random effects model with robust standard errors is used, adopting different definitions, which expand the range to the top 20 % and the top 30 %. As shown in Table 8, all test results remain consistent, confirming support for all hypotheses except Hypothesis 3b. Regarding Hypotheses 2a and 2b for testing the moderation of corporate reputation on the relationship between product popularity and digital video game sales, their predictive margin plots also show significant marginal differences if the definition of reputable firms is extended to the top 20 or top 30 percent of the video game companies (Fig. 4).

Table 8
Robustness check with top 20 and top 30 percent companies.

Dependent: Ln (Daily Sales $_{i,t}$)	**Top20**			**Top30**		
Corporate Reputation: *Total Revenue of Company*	**Coeff.**	**S. E.**	**p-value**	**Coeff.**	**S. E.**	**p-value**
Constant	8.9222	0.0220	0.01 >	8.9108	0.0219	0.01 >
Developer Reputation $_{i,t-1}$	0.2231	0.1023	0.03	0.2541	0.1002	0.01
Publisher Reputation $_{i,t-1}$	0.3358	0.0612	0.01 >	0.3248	0.0528	0.01 >
Product Popularity $_{i,t-1}$	−0.0006	0.0001	0.01 >	−0.0005	0.0001	0.01 >
Developer Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >	−0.0004	0.0001	0.01 >
Publisher Reputation $_{i,t-1}$**Product Popularity* $_{i,t-1}$	−0.0003	0.0001	0.01 >	−0.0003	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0471	0.0220	0.03	0.0471	0.0220	0.03
Publisher Coopetitive Branding $_i$	−0.0154	0.0333	0.64	−0.0148	0.0333	0.66

Fourth, an additional analysis is performed by extending the lagged time to t-2 to mitigate further the possible correlation between the predictors and the error term, lessening the concern about endogeneity. After extending the time lag to t-2, 110,779 observations of 15,450 digital video games are used in the test, discarding 200,871 observations from the initial data. This robustness check is performed using a random effects model with robust standard errors, which was adopted for the main analysis. The results align with the main analysis results, although the coefficients and p-values of two terms related to publisher reputation—*Publisher Reputation* $_{i,t-2}$ and *Publisher Reputation* $_{i,t-2}$**Product Popularity* $_{i,t-2}$—slightly vary by the definitions (Table 9).

Fifth, corporate reputation can be determined by not only overall financial performance (i.e., total revenue) but also constant product quality (Charters, 2009). It is possible to build and maintain a strong reputation through consistently satisfying consumers with a few best-selling products. Despite having relatively low total revenue, such companies can still be considered reputable in a market. To address this possibility, the median revenue per video game is employed as an alternative variable for *Developer Reputation* and *Publisher Reputation* in the robustness check. A random effects model with robust standard errors adopted for the main analysis is applied to this dataset to test hypotheses. Hypothesis test results using the alternative variable are consistent with those from the main analysis. All the test results closely mirror those from the main analysis when expanding the definition to the top 20 % and 30 %. Therefore, all the hypotheses are supported, except for Hypothesis 3b (Table 10).

Lastly, accumulated product rating scores can represent corporate reputation (Gatti et al., 2012) because the perceived quality of individual products is closely related to beliefs about the brand and brand favorability (Lange et al., 2011), playing as a cue to signal the firm reputation in the market (Rindova et al., 2005). Thus, each company's average video game rating is adopted as an alternative variable for *Developer Reputation* and *Publisher Reputation* for the robustness check. An additional data collection was conducted from steambase.io, which provides each video game's "Player Score". The average video game rating of developers or publishers was estimated based on the player score and applied to the research model using a random effects model with robust standard errors. The overall hypothesis test results remain constant with those from the tests adopting revenue-based reputation measures (Table 11). Two noticeable differences are that *Developer Reputation* $_{i,t-1}$**Product Popularity* $_{i,t-1}$ is not significant ($p = 0.22$), and *Developer Coopetitive Branding* $_i$ is marginally significant at the 90 % confidence level ($p = 0.07$) when reputable developers are defined as the top 30 % of firms based on average video game ratings.

Given all the analysis results, it is concluded that all hypotheses are supported except for Hypothesis 3b, as summarized in Table 12.

6. Discussion and contributions

6.1. Discussion

This study represents one of the first efforts to investigate the distinct impacts of developer and publisher corporate reputations on digital video game sales, utilizing a comprehensive dataset that combines both upstream and downstream sectors of the digital video game value chain. Additionally, it explored the interaction of their reputation on the relationship between product popularity and sales and the impact of coopetitive branding of developers and publishers based on the theoretical framework of signaling theory. Overall, the findings shed light and expand the ideas of signaling theory, except for the insignificant relationship between publisher coopetitive branding and digital video game sales found in the result of Hypothesis 3b.

In support of Hypotheses 1a and 1b, the study reveals a positive

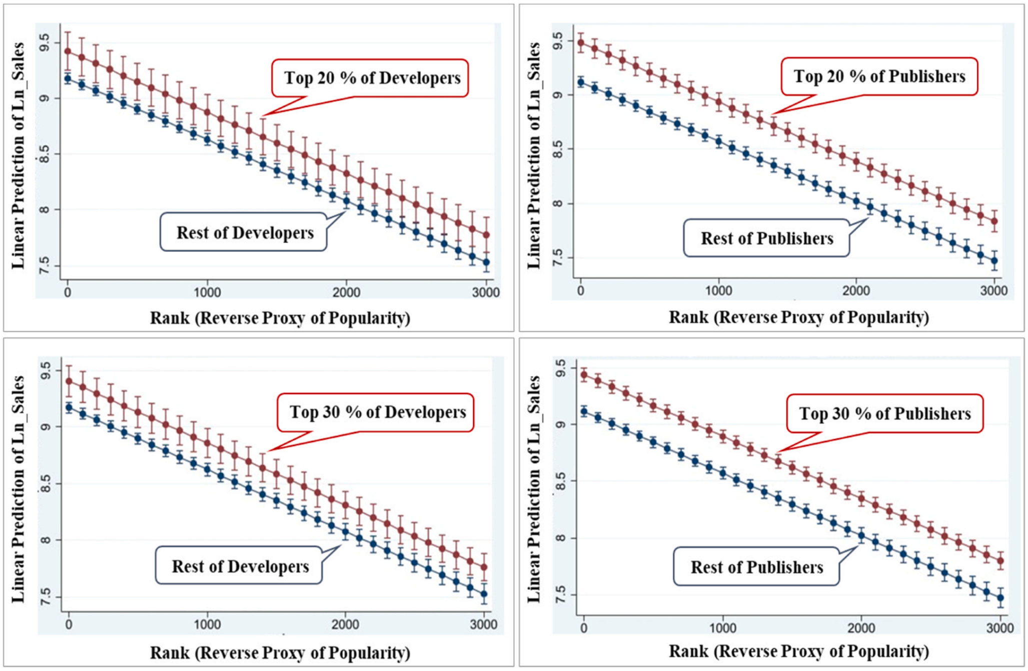

Fig. 4. Predictive margin plots: Moderation of developer and publisher reputation (Top 20 % & 30 %).

Table 9
Robustness check with lagged time effect (t-2).

Dependent: Ln (Daily Sales $_{i,t}$)	**Top10**			**Top20**			**Top30**		
Corporate Reputation: *Total Revenue of Company*	**Coeff.**	**S.E.**	**p-value**	**Coeff.**	**S.E.**	**p-value**	**Coeff.**	**S.E.**	**p-value**
Constant	8.9937	0.0264	0.01 >	8.9805	0.0266	0.01 >	8.9766	0.0268	0.01 >
Developer Reputation $_{i,t-2}$	0.1509	0.0522	0.01 >	0.1701	0.0455	0.01 >	0.1732	0.0425	0.01 >
Publisher Reputation $_{i,t-2}$	0.0391	0.0365	0.28	0.0751	0.0345	0.03	0.0582	0.0341	0.09
Product Popularity $_{i,t-2}$	−0.0007	0.0001	0.01 >	−0.0007	0.0001	0.01 >	−0.0007	0.0001	0.01 >
Developer Reputation $_{i,t-2}$ **Product Popularity* $_{i,t-2}$	−0.0002	0.0001	0.01 >	−0.0002	0.0001	0.01 >	−0.0002	0.0001	0.01 >
Publisher Reputation $_{i,t-2}$ **Product Popularity* $_{i,t-2}$	−0.0002	0.0220	0.01 >	−0.0002	0.0001	0.01 >	−0.0001	0.0001	0.14
Developer Coopetitive Branding $_i$	0.0440	0.0216	0.04	0.0428	0.0216	0.04	0.0407	0.0216	0.05
Publisher Coopetitive Branding $_i$	−0.0354	0.0246	0.15	−0.0301	0.0245	0.22	−0.0265	0.0245	0.27

Table 10
Robustness check with median revenue per game.

Dependent: Ln (Daily Sales $_{i,t}$)	**Top10**			**Top20**			**Top30**		
Corporate Reputation: *Median Revenue*	**Coeff.**	**S.E.**	**p-value**	**Coeff.**	**S.E.**	**p-value**	**Coeff.**	**S.E.**	**p-value**
Constant	8.9656	0.0164	0.01 >	8.9669	0.0164	0.01 >	8.9546	0.0164	0.01 >
Developer Reputation $_{i,t-2}$	1.2544	0.1569	0.01 >	0.6216	0.0942	0.01 >	0.4067	0.0555	0.01 >
Publisher Reputation $_{i,t-2}$	0.3540	0.1403	0.01	0.5234	0.0877	0.01 >	0.5623	0.0447	0.01 >
Product Popularity $_{i,t-2}$	−0.0006	0.0000	0.01 >	−0.0006	0.0000	0.01 >	−0.0006	0.0000	0.01 >
Developer Reputation $_{i,t-2}$ **Product Popularity* $_{i,t-2}$	−0.0016	0.0003	0.01 >	−0.0009	0.0002	0.01 >	−0.0005	0.0001	0.01 >
Publisher Reputation $_{i,t-2}$ **Product Popularity* $_{i,t-2}$	−0.0003	0.0002	0.09	−0.0006	0.0001	0.01 >	−0.0008	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0523	0.0163	0.01 >	0.0541	0.0163	0.01 >	0.0561	0.0163	0.01 >
Publisher Coopetitive Branding $_i$	−0.0256	0.0186	0.16	−0.0212	0.0186	0.25	−0.0113	0.0186	0.54

impact of developers' and publishers' corporate reputation on digital video game sales, consistently validated across various robustness checks. These findings align with signaling theory, which explains consumer purchase behavior in a competitive market (Shapiro, 1983), such as the digital video game market. They also align with the literature emphasizing the importance of brand reputation for experience products (Grunwald & Hempelmann, 2010).

Based on the support from Hypotheses 2a and 2b, the reputation of both developers and publishers enhances the positive relationship between product popularity and digital video game sales. The findings resonate with the brand management literature, claiming that the effect of perceived product popularity can be more significant if the product

Table 11
Robustness check with average video game rating of companies.

Dependent: Ln (Daily Sales $_{i,t}$)	Top10			Top20			Top30		
Corporate Reputation: *Average game rating*	**Coeff.**	**S.E.**	**p-value**	**Coeff.**	**S.E.**	**p-value**	**Coeff.**	**S.E.**	**p-value**
Constant	8.8851	0.0219	0.01 >	8.8343	0.0222	0.01	8.7888	0.0232	0.01 >
Developer Reputation $_{i,t-2}$	0.2958	0.0637	0.01 >	0.1264	0.0514	0.01 >	0.1413	0.0453	0.01 >
Publisher Reputation $_{i,t-2}$	0.2310	0.0625	0.01	0.3516	0.0460	0.01 >	0.3068	0.0446	0.01 >
Product Popularity $_{i,t-2}$	−0.0005	0.0001	0.01 >	−0.0005	0.0001	0.01 >	−0.0005	0.0001	0.01 >
Developer Reputation $_{i,t-2}$ **Product Popularity* $_{i,t-2}$	−0.0002	0.0001	0.01 >	−0.0001	0.0001	0.03	0.0001	0.0001	0.22
Publisher Reputation $_{i,t-2}$ **Product Popularity* $_{i,t-2}$	−0.0001	0.0001	0.07	−0.0002	0.0001	0.01 >	−0.0002	0.0001	0.01 >
Developer Coopetitive Branding $_i$	0.0520	0.0219	0.02	0.0504	0.0219	0.02	0.0396	0.0219	0.07
Publisher Coopetitive Branding $_i$	−0.0360	0.0333	0.27	−0.0443	0.0332	0.18	−0.0378	0.0331	0.25

Table 12
Summary of hypothesis test results.

Hypothesis		Result
H1a	Developer reputation has a positive relationship with digital video game sales.	*Supported*
H1b	Publisher reputation has a positive relationship with digital video game sales.	*Supported*
H2a	The positive relationship between product popularity and sales is larger for digital video games of reputable developers.	*Supported*
H2b	The positive relationship between product popularity and sales is larger for digital video games of reputable publishers.	*Supported*
H3a	Developer coopetitive branding has a positive relationship with digital video game sales.	*Supported*
H3b	Publisher coopetitive branding has a positive relationship with digital video game sales.	*Not Supported*

has additional strengths in its corporate brand (Hirvonen et al., 2013; Pecot et al., 2018). The interaction effect of developers and publisher reputation is constantly significant when the reputation is operationalized as the total company revenue, median revenue per video game, and average rating of video games. This implies that consumers of digital video games consider overall firm performance, the success of individual video games, and perceived quality of other video games, which are operationalized in this study as total revenue, median revenue per game, and average rating of video games. As key quality cues, they can aid consumers in lowering information asymmetry and uncertainty toward product quality.

Finally, the study argues that branding, often overlooked as a quality cue, is a rich source of information for consumers. The findings reveal that the coopetitive branding of developers significantly impacts digital video game sales, while the same effect is not observed for publishers, as indicated by the test results of Hypothesis 3a and 3b. This is interesting because, despite the general trend of increased brand quality with more brand alliances (Fang et al., 2013), the effect does not hold for coopetitive branding among publishers.

This unexpected finding may be explained by collaboration theory, which asserts that the productivity of collaboration between business organizations depends on task attributes, including whether the task is independent or tightly coupled, and time-sensitive or non-time-sensitive (Patel et al., 2012). Compared to video game development, publishing pertains to more tightly coupled and time-sensitive tasks, such as user complaint management, content update, and system patch, which require close collaboration among the teams responsible for the tasks. However, these collaborative efforts often encounter difficulties when occurring among multiple organizations: misunderstandings, miscommunications, and conflicts due to different organizational cultures (Patel et al., 2012). Consequently, the quality of collaboration outcomes may not significantly differ from that of a single organization, and consumers would not perceive the involvement of multiple publishers as an additional advantage. This also resonates with the idea that inter-organizational collaboration for disaster responses or emergency plans, which are highly tangled and time-sensitive, generates few advantages due to such issues (Abbas & Miller, 2025). Another possible explanation is the operational independence of co-publishers. In many cases, publishers function autonomously with limited collaboration, which hinder additional value creation for their consumers. For instance, when launching video games in foreign markets, publishers often delegate responsibilities to local partners due to legal constraints and language barriers (Kuhns, 2020). In such cases, the potential benefits of a coopetitive branding strategy among publishers could be negligible from the consumer's perspective, as partnered publishers abroad are not directly involved in service quality within the local market.

6.2. Theoretical implications

This study offers novel contributions to the literature on signaling theory with the expansion and inclusion of new product quality cues, with evidence from digital video game sales. First, for insights into reputation as a quality cue, this is among the initial studies to thoroughly investigate the impact of developers' and publishers' corporate reputations on digital video game sales. Even though there were a few prior studies, they yielded contradictory findings, potentially due to their inconsistent and problematic definitions of corporate reputation (Choi et al., 2018; Cox, 2014). Through a refined definition of reputation and comprehensive analysis accounting for multiple dimensions of corporate reputation, this study clarifies the significant impact of both developers' and publishers' reputations on digital video game sales. This supports that corporate reputation serves as a signal of video game's potential quality, influencing consumer purchasing decisions on e-commerce platforms while challenging previous research that downplayed the impact of corporate reputation on video game sales (Choi et al., 2018; Gil & Warzynski, 2015; Nam & Kim, 2018; Yi et al., 2019).

Second, the prior studies examining the impact of brand reputation on product sales mostly center on traditional markets, such as hotels (Anggani & Suherlan, 2020), restaurants (Han et al., 2011a), fashion products (Lee & Lee, 2018), and fast-moving consumer goods (Huang & Sarigöllü, 2012). The findings of this study illustrate that consumers perceive the corporate reputations of both developers and publishers as signals of potential game quality, extending the effect of brand reputation to the IT domains.

Third, extant signaling theory research on e-commerce has mainly focused on the direct effect of quality cues on product sales, such as website quality (Wells et al., 2011), product reviews (Chen et al., 2020), seller reputation (Al-Adwan & Yaseen, 2023), real-time communication with sellers (Dong et al., 2024), product recommendation (Li et al., 2024), and trust badges for online sellers (Lladós-Masllorens et al., 2020). The findings of this study reveal that corporate reputation amplifies the effect of product popularity on sales, suggesting the interaction between quality cues. Developers' and publishers' reputations enhance the impact of popularity on digital video game sales. This finding also contributes to the literature on corporate brand strategy, emphasizing the financial benefits of a well-established reputation (Chierici et al., 2019; Sun et al., 2024) by illustrating the positive moderation of corporate reputation on the relationship between product popularity and sales.

Fourth, this study constitutes one of the earliest efforts to distinguish and compare the effects of developer and publisher reputations—treated as distinct quality signals for mitigating information asymmetry—on video game sales. The findings regarding this differentiation suggest that developer reputation exerts a far greater influence on consumer purchase decisions than publisher reputation. By demonstrating that quality signals originating from different entities within the value chain can differentially affect product market performance based on their perceived roles, this study broadens the theoretical scope of signaling theory in the extant literature (Erdem & Swait, 2001; Pecot et al., 2018; Wesley II et al., 2022).

Lastly, while a coopetitive branding strategy, serving as an additional quality cue, at the upstream stage (e.g., product development) effectively attracts consumers and boosts product sales, its implementation at the downstream (e.g., product distribution and marketing) appears to have a minimal impact. This finding contributes to the video game literature based on signaling theory (Choi et al., 2018; Rodrigues et al., 2024; Tang & He, 2022), as well as to the business strategy literature on coopetition (Hameed & Naveed, 2019). It suggests that the involvement of multiple developers may be perceived as a signal for superior quality—introducing a novel quality cue in the context of video games—whereas the presence of multiple publishers may not convey the same effect.

6.3. Strategic implications

Practitioners in software industries, including the digital video game sector, can utilize the findings of this study to develop strategies for enhancing their products' market performance. Reputable companies—both developers and publishers—should emphasize their corporate brands through diverse marketing campaigns. These efforts may include in-game introductions, advertising, public relations, and point-of-sale information on sales platforms. This can be particularly crucial when their video games gain popularity, as corporate reputation can amplify the impact of product popularity on digital video game sales. Therefore, practitioners may consider focusing marketing campaigns on popular video games that highlight corporate brands of both developers and publishers.

The findings of this study provide practical suggestions for both small, indie companies and large corporations in the video game industry. The study's findings can be helpful for indie developers who hesitate to collaborate with established publishers due to the contradictory arguments regarding the collaboration (Mendez, 2017). Given the significant impact of publisher reputation, indie developers lacking corporate reputation should consider partnering with established publishers, as their reputation plays a pivotal role in digital video game sales. Particularly, indie developers without market presence should consider this approach, as their video games may struggle to gain visibility without reputable publishers, especially in the highly competitive digital video game market, where over 10,000 new titles are released annually (Statista, 2025). However, indie developers with highly rated game titles may not need to partner with large publishers, as these successful titles serve as quality signals that positively influence video game sales. Moreover, the study' results indicate that developer reputation has a greater impact than publisher reputation on video game sales. These findings recommend that indie developers with quality titles can be more confident when selecting their publishing partners, even if they do not yet generate substantial revenue.

For large companies in the video game domain, the findings suggest that established developers may not need to align with major publishers, as developer reputation is more critical in driving video game sales than publisher reputation. This insight implies that established developers could negotiate for a large share of profits and greater creative control in video game development, even when partnering with large publishers. Nonetheless, publisher reputation still has a positive impact on sales, as well as enhancing the effect of video game popularity on sales. Therefore, when all other conditions are comparable, partnering with reputable publishers remains advantageous, even for well-regarded developers in terms of revenue and perceived quality of game titles. Conversely, these findings suggest that reputable publishers should evaluate potential developer partners based not only on total revenue but also on the current quality and ratings of their games. As a result, publishers may find greater value in partnering with smaller developers that boast a strong game portfolio, rather than larger ones with revenue alone.

Lastly, for practitioners who consider developing marketing strategies for video games involving multiple partners, the findings suggest that coopetitive branding is an effective strategy for developers but not necessarily for publishers. When handling digital video games involving multiple developers, practitioners should employ a coopetitive branding strategy, highlighting the involvement of these developers in their campaigns. This supports the growing popularity of coopetitive development in the IT domains (Zapadka et al., 2022), including the video game industry (Agate, 2023), encouraging developers not to view their competitors negatively but to adopt this approach for their market success. However, coopetitive branding for publishers may not significantly impact sales. Therefore, featuring multiple publishers in marketing communications might not be essential, although they collaborate at the downstream.

7. Limitations and future research

While this study contributes valuable insights, it also has limitations that open up opportunities for future research. Although this study meticulously considered various operational measures for developers' and publishers' reputations, it could not encompass all dimensions of reputation, such as awareness, likability, trust, and overall perception (Van Der Merwe & Puth, 2014; Walker, 2010). Since these perceptual dimensions cannot be captured through corporate financial metrics and consumer ratings but are instead reflected in consumer perceptions, this study could not examine them. Future research could employ alternative methodologies—such as surveys or interviews—to examine how various dimensions of corporate reputation function differently as quality signals.

In operationalizing coopetitive branding, this study utilized a binary variable, which, while capturing the core concept—namely, the joint presentation of two or more independent brands from competing firms on a single product (Chiambaretto et al., 2016)—does not account for the depth, strength, or quality of the collaboration. Moreover, the effectiveness of a coopetitive branding strategy may depend on the success of prior collaborations. For example, consumers may perceive publisher coopetitive branding as a strong quality signal when the partnering firms have a demonstrated history of successful joint projects (or conversely, as a negative signal in the presence of a failed collaboration history). However, due to data limitations regarding previous collaboration outcomes, this study was unable to account for such effects, which may limit the theoretical generalizability of its findings regarding signaling theory. Future research could explore more nuanced dimensions of coopetitive branding to assess how its impact varies depending on factors not captured in the present analysis.

Another limitation pertains to the data sources used in this study. While integrating data from multiple sources, this study had to exclude sales data for certain video games due to missing developer or publisher information and their low daily unit sales, potentially introducing sample selection bias. In particular, video games developed or published by small companies were particularly susceptible to this issue.

Additionally, the data covers only video games utilizing Steampowered.com as the sales platform. Although Steam is the predominant sales platform for digital video games, other platforms, such as store.epicgames.com (Epic Games, Inc.) and gog.com (CD Projekt), exist. Researchers studying corporate reputation dynamics in the digital video game market may consider other platforms to assess the generalizability of this study's findings across different contexts.

CRediT authorship contribution statement

Hoon S. Choi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Emmanuel W. Ayaburi:** Conceptualization, Validation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Abbas, R., & Miller, T. (2025). Exploring communication inefficiencies in disaster response: Perspectives of emergency managers and health professionals. *International Journal of Disaster Risk Reduction, 120*, Article 105393.

Adair, C. (2023). Video game addiction statistics 2023 – how many addicted gamers are there? Game Quitters.

Agate. (2023). Co-Development" the Future of Video Game Production. Agate, agate.id.

Aiken, K. D., & Boush, D. M. (2006). Trustmarks, objective-source ratings, and implied investments in advertising: Investigating online trust and the context-specific nature of Internet signals. *Journal of the Academy of Marketing Science, 34*, 308–323.

Al-Adwan, A. S., & Yaseen, H. (2023). Solving the product uncertainty hurdle in social commerce: The mediating role of seller uncertainty. *International Journal of Information Management Data Insights, 3*, Article 100169.

Anggani, M. F., & Suherlan, H. (2020). E-reputation management of hotel industry. *Journal of Hospitality Tourism Education*, 2–17.

Azzari, V., & Pelissari, A. (2021). Does brand awareness influences purchase intention? The mediation role of brand equity dimensions. *Bbr Brazilian Business Review, 17*, 669–685.

Babb, J., & Terry, N. (2013). Comparing video game sales by gaming platform. *Southwestern Economic Review, 40*, 25–46.

Bahar, V. S., Nenonen, S., & Starr Jr, R. G. (2022). On the same boat but singing a different tune: Coopetition between hotels and platforms close to customers. *Industrial Marketing Management, 107*, 52–69.

Bardhan, I., Krishnan, V., & Lin, S. (2013). Research note—business value of information technology: Testing the interaction effect of IT and R&D on tobin's q. *Information Systems Research, 24*, 1147–1161.

Barnett, M. L., Jermier, J. M., & Lafferty, B. A. (2006). Corporate reputation: The definitional landscape. *Corporate Reputation Review, 9*, 26–38.

Binken, J. L., & Stremersch, S. (2009). The effect of superstar software on hardware sales in system markets. *Journal of Marketing, 73*, 88–104.

Bouncken, R. B., Fredrich, V., Ritala, P., & Kraus, S. (2018). Coopetition in new product development alliances: Advantages and tensions for incremental and radical innovation. *British Journal of Management, 29*, 391–410.

Bray, S. (2024). How many new games are on Steam a day? NCESC.COM, ncesc.com.

Calvin, A. (2018). Ubisoft partners with Tencent to release games in China. Pocket Gamer, pocketgamer.biz.

Chai, L., Li, J., Tangpong, C., & Clauss, T. (2020). The interplays of coopetition, conflicts, trust, and efficiency process innovation in vertical B2B relationships. *Industrial Marketing Management, 85*, 269–280.

Charters, S. (2009). Does a brand have to be consistent? *Journal of Product Brand Management, 18*, 284–291.

Chen, Y. F. (2008). Herd behavior in purchasing books online. *Computers in Human Behavior, 24*, 1977–1992.

Chen, L., Baird, A., & Straub, D. (2020). A linguistic signaling model of social support exchange in online health communities. *Decision Support Systems, 130*, Article 113233.

Chen, Y., Lu, Y., Wang, B., & Pan, Z. (2019). How do product recommendations affect impulse buying? An empirical study on WeChat social commerce. *Information Management, 56*, 236–248.

Chen, Y. C., Shang, R. A., & Kao, C. Y. (2009). The effects of information overload on consumers' subjective state towards buying decision in the Internet shopping environment. *Electronic Commerce Research and Applications, 8*, 48–58.

Cheung, C. M., Xiao, B. S., & Liu, I. L. (2014). Do actions speak louder than voices? The signaling role of social information cues in influencing consumer purchase decisions. *Decision Support Systems, 65*, 50–58.

Chiambaretto, P., Gurău, C., & Le Roy, F. (2016). Coopetitive branding: definition, typology, benefits and risks. *Industrial Marketing Management, 57*, 86–96.

Chierici, R., Del Bosco, B., Mazzucchelli, A., & Chiacchierini, C. (2019). Enhancing brand awareness, reputation and loyalty: The role of social media. *International Journal of Business and Management, 14*, 216–228.

Choi, H. S., & Chen, C. (2019). The effects of discount pricing and bundling on the sales of game as a service: an empirical investigation. *Journal of Electronic Commerce Research, 20*, 21–34.

Choi, H. S., Ko, M. S., Medlin, D., & Chen, C. (2018). The effect of intrinsic and extrinsic quality cues of digital video games on sales: An empirical investigation. *Decision Support Systems, 106*, 86–96.

Chun, R. (2005). Corporate reputation: Meaning and measurement. *International Journal of Management Reviews, 7*, 91–109.

Chun, R., Da Silva, R., Davies, G., & Roper, S. (2005). *Corporate reputation and competitiveness*. Routledge.

Corts, K. S., & Lederman, M. (2009). Software exclusivity and the scope of indirect network effects in the US home video game market. *International Journal of Industrial Organization, 27*, 121–136.

Cowan, K., & Guzman, F. (2020). How CSR reputation, sustainability signals, and country-of-origin sustainability reputation contribute to corporate brand performance: an exploratory study. *Journal of Business Research, 117*, 683–693.

Cox, J. (2014). What makes a blockbuster video game? An empirical analysis of US sales data. *Managerial and Decision Economics, 35*, 189–198.

Cretu, A. E., & Brodie, R. J. (2007). The influence of brand image and company reputation where manufacturers market to small firms: A customer value perspective. *Industrial Marketing Management, 36*, 230–240.

Crick, J. M., Friske, W., & Morgan, T. A. (2024). The relationship between coopetition strategies and company performance under different levels of competitive intensity, market dynamism, and technological turbulence. *Industrial Marketing Management, 118*, 56–77.

Dawar, N., & Parker, P. (1994). Marketing universals: Consumers' use of brand name, price, physical appearance, and retailer reputation as signals of product quality. *Journal of Marketing, 58*, 81–95.

Doh, J. P., Howton, S. D., Howton, S. W., & Siegel, D. S. (2010). Does the market respond to an endorsement of social responsibility? The role of institutions, information, and legitimacy. *Journal of Management, 36*, 1461–1485.

Dong, L., Hua, Z., Huang, L., Ji, T., Jiang, F., Tan, G., & Zhang, J. (2024). The impacts of live chat on service–product purchase: Evidence from a large online outsourcing platform. *Information Management, 61*, Article 103931.

Duan, W., Gu, B., & Whinston, A. B. (2009). Informational cascades and software adoption on the Internet: an empirical investigation. *MIS Quarterly*, 23–48.

Eccles, R. G., Newquist, S. C., & Schatz, R. (2007). Reputation and its risks. *Harvard Business Review, 85*, 104.

Erdem, T., & Swait, J. (2001). Brand equity as a signaling. *Journal of Consumer Psychology, 7*, 131–157.

Erdil, K., Finn, E., Keating, K., Meattle, J., Park, S., & Yoon, D. (2003). Software maintenance as part of the software life cycle. *Comp180: Software Engineering Project, 1*, 1–49.

Fang, X., Gammoh, B. S., & Voss, K. E. (2013). Building brands through brand alliances: Combining warranty information with a brand ally. *Journal of Product Brand Management, 22*, 153–160.

Gatti, L., Caruana, A., & Snehota, I. (2012). The role of corporate social responsibility, perceived quality and corporate reputation on purchase intention: implications for brand management. *Journal of Brand Management, 20*, 65–76.

Gernsheimer, O., Kanbach, D. K., & Gast, J. (2021). Coopetition research-A systematic literature review on recent accomplishments and trajectories. *Industrial Marketing Management, 96*, 113–134.

Gil, R., & Warzynski, F. (2015). Vertical integration, exclusivity, and game sales performance in the US video game industry. *The Journal of Law, Economics, and Organization, 31*, 143–168.

Gilbert, D. (2015). Let's talk steam spy. *Wadjeteye Games*. Wadjeteye Games.

GMG. (2024). We're Green Man Gaming. Green Man Gaming, greenmangaming.com.

Greene, W. H. (2003). *Econometric analysis*. India: Pearson Education.

Gregg, D. G., & Walczak, S. (2008). Dressing your online auction business for success: An experiment comparing two ebay businesses. *MIS Quarterly*, 653–670.

Grunwald, G., & Hempelmann, B. (2010). Impacts of reputation for quality on perceptions of company responsibility and product-related dangers in times of product-recall and public complaints crises: Results from an empirical investigation. *Corporate Reputation Review, 13*, 264–283.

Hameed, W. U., & Naveed, F. (2019). Coopetition-based open-innovation and innovation performance: Role of trust and dependency evidence from Malaysian high-tech SMEs. *Pakistan Journal of Commerce and Social Sciences (PJCSS, 13*, 209–230.

Han, K., Kauffman, R. J., & Nault, B. R. (2011b). Research note—Returns to information technology outsourcing. *Information Systems Research, 22*, 824–840.

Han, H., Kim, W., & Hyun, S. S. (2011a). Switching intention model development: Role of service performances, customer satisfaction, and switching barriers in the hotel industry. *International Journal of Hospitality Management, 30*, 619–629.

Hausman, J. A., & Taylor, W. E. (1981). Panel data and unobservable individual effects. *Econometrica: Journal of the Econometric Society*, 1377–1398.

Hirvonen, S., Laukkanen, T., & Reijonen, H. (2013). The brand orientation-performance relationship: An examination of moderation effects. *Journal of Brand Management, 20*, 623–641.

Hong, J. F., & Snell, R. S. (2015). Knowledge development through co-opetition: A case study of a Japanese foreign subsidiary and its local suppliers. *Journal of World Business, 50*, 769–780.

Huang, R., & Sarigöllü, E. (2012). How brand awareness relates to market outcome, brand equity, and the marketing mix. *Journal of Business Research, 65*, 92–99.

Hur, W.-M., Kim, H., & Woo, J. (2014). How CSR leads to corporate brand equity: Mediating mechanisms of corporate brand credibility and reputation. *Journal of Business Ethics, 125*, 75–86.

Ifrach, B., Johari, R. (2014) The impact of visibility on demand in the market for mobile apps. Available at SSRN 2444542.

Jensen, M., & Roy, A. (2008). Staging exchange partner choices: When do status and reputation matter? *Academy of Management Journal, 51*, 495–516.

Jeong, H. J., & Kwon, K. N. (2012). The effectiveness of two online persuasion claims: Limited product availability and product popularity. *Journal of Promotion Management, 18*, 83–99.

Jung, N. Y., & Seock, Y.-K. (2016). The impact of corporate reputation on brand attitude and purchase intention. *Fashion and Textiles, 3*, 1–15.

Kathuria, A., Karhade, P. P., Ning, X., & Konsynski, B. R. (2023). Blood and water: information technology investment and control in family-owned businesses. *Journal of Management Information Systems, 40*, 208–238.

Klimas, P., & Czakon, W. (2018). Organizational innovativeness and coopetition: A study of video game developers. *Review of Managerial Science, 12*, 469–497.

Koutsou-Wehling, N. (2024). Video Games: Sales, Revenue, Companies, Regions & Development. ECDB, ecommercedb.com.

Kuhns, T. (2020). How To Publish Your Game In China. AppInChina, AppInChina.co.

Landsman, V., & Stremersch, S. (2011). Multihoming in two-sided markets: An empirical inquiry in the video game console industry. *Journal of Marketing, 75*, 39–54.

Lange, D., Lee, P. M., & Dai, Y. (2011). Organizational reputation: A review. *Journal of Management, 37*, 153–184.

Lee, J., & Lee, Y. (2018). Effects of multi-brand company's CSR activities on purchase intention through a mediating role of corporate image and brand image. *Journal of Fashion Marketing and Management: An International Journal, 22*, 387–403.

Lee, G., & Raghu, T. S. (2014). Determinants of mobile apps' success: Evidence from the app store market. *Journal of Management Information Systems, 31*, 133–170.

Lewis, S. (2001). Measuring corporate reputation. *Corporate Communications: An International Journal, 6*, 31–35.

Li, X., Wang, K., & Jiang, Q. (2024). How to make recommendations on mobile social E-commerce more effective: The role of social features and temporal cues. *Information Management*, Article 104002.

Lioliou, E., Krancher, O., & Oshri, I. (2025). The complementary and substitutional effects of forced and emergent mechanisms in multisourcing. *The Journal of Strategic Information Systems, 34*, Article 101869.

Liu, C. Z., Au, Y. A., & Choi, H. S. (2014). Effects of freemium strategy in the mobile app market: An empirical study of google play. *Journal of Management Information Systems, 31*, 326–354.

Liu, M., & Xiong, W. (2023). Well-known or well-liked? the effects of brand reputation on customers' responses to service failure. *Journal of Travel Tourism Marketing, 40*, 221–241.

Lladós-Masllorens, J., Meseguer-Artola, A., & Rodríguez-Ardura, I. (2020). Understanding peer-to-peer, two-sided digital marketplaces: Pricing lessons from airbnb in barcelona. *Sustainability, 12*, 5229.

Logan, D. (2018). Publishing 101 — Should You Partner with a Game Publisher? Medium, akuparagames.medium.com.

Lu, B., & Chen, Z. (2021). Live streaming commerce and consumers' purchase intention: An uncertainty reduction perspective. *Information Management, 58*, Article 103509.

Mallapragada, G., Chandukala, S. R., & Liu, Q. (2016). Exploring the effects of "What"(product) and "Where"(website) characteristics on online shopping behavior. *Journal of Marketing, 80*, 21–38.

Malvankar, K., Fallon, E., Connolly, P., & Flanagan, K. (2023). *A case study to analyze the impact of social media on video game sales, 2023 3rd international conference on intelligent communication and computational techniques (ICCT)* (pp. 1–5). IEEE.

Mansfield, E. R., & Helms, B. P. (1982). Detecting multicollinearity. *The American Statistician, 36*, 158–160.

Mavlanova, T., Benbunan-Fich, R., & Koufaris, M. (2012). Signaling theory and information asymmetry in online commerce. *Information Management, 49*, 240–247.

Mavlanova, T., Benbunan-Fich, R., & Lang, G. (2016). The role of external and internal signals in E-commerce. *Decision Support Systems, 87*, 59–68.

Mendez, J. (2017). Blackshellmedia.com/. *Should You Self-Publish or Work With a Publisher for Your Next Indie Game? Black Shell Media*.

Mirocha, J., Bents, R., LaBrosse, M., & Rietow, K. (2013). Strategies for developing leaders in small to medium sized firms: An analysis of best practices in the most successful firms. *Organization Development Journal, 31*.

Mishra, S., Singh, S. N., Fang, X., & Yin, B. (2017). Impact of diversity, quality and number of brand alliance partners on the perceived quality of a new brand. *Journal of Product Brand Management, 26*, 159–176.

Mróz-Gorgoń, B. (2016). Co-branding as a strategy-fashion market perspective: Prestige or masstige? *International Journal of Sales, Retailing Marketing, 5*.

Nam, K., Kim, H. (2018). Predicting success of mobile game after market launch using app downloads and sales revenues as performance indicators, 29th European Regional Conference of the International Telecommunications Society (ITS). International Telecommunications Society (ITS), Trento, Italy.

Naser, K., Nuseibeh, R., & Al-Hadeya, A. (2013). Factors influencing corporate working capital management: Evidence from an emerging economy. *Journal of Contemporary Issues in Business Research, 2*, 11–30.

Nieto, M. J., & Santamaría, L. (2007). The importance of diverse collaborative networks for the novelty of product innovation. *Technovation, 27*, 367–377.

Nishant, R., Nguyen, T., Teo, T. S., & Hsu, P.-F. (2023). Role of substantive and rhetorical signals in the market reaction to announcements on AI adoption: A configurational study. *European Journal of Information Systems*, 1–43.

Obedkov, E. (2023). Over 14k games released on steam in 2023: Deep dive into most played titles, record numbers, and major failures. *Game World Observer*.

Oh, Y. K., & Min, J. (2015). The mediating role of popularity rank on the relationship between advertising and in-app purchase sales in mobile application market. *Journal of Applied Business Research (JABR), 31*, 1311–1322.

Ou, C. X., & Chan, K. C. (2014). Developing a competitive edge in electronic markets via institutional and social based quality signaling mechanisms. *Information Management, 51*, 532–540.

Park, B.-J. R., Srivastava, M. K., & Gnyawali, D. R. (2014). Walking the tight rope of coopetition: Impact of competition and cooperation intensities and balance on firm innovation performance. *Industrial Marketing Management, 43*, 210–221.

Patel, H., Pettitt, M., & Wilson, J. R. (2012). Factors of collaborative working: A framework for a collaboration model. *Applied Ergonomics, 43*, 1–26.

Pecot, F., Merchant, A., Valette-Florence, P., & De Barnier, V. (2018). Cognitive outcomes of brand heritage: A signaling perspective. *Journal of Business Research, 85*, 304–316.

Quintana-Garcia, C., & Benavides-Velasco, C. A. (2004). Cooperation, competition, and innovative capability: A panel data of european dedicated biotechnology firms. *Technovation, 24*, 927–938.

Rarick, C. A., & Nickerson, I. (2006). An empirical analysis of web-based corporate mission statements. *Journal of Strategic E-Commerce, 4*(1).

Readman, J., & Grantham, A. (2006). Shopping for buyers of product development expertise: How video games developers stay ahead. *European Management Journal, 24*, 256–269.

Rhee, M., & Valdez, M. E. (2009). Contextual factors surrounding reputation damage with potential implications for reputation repair. *Academy of Management Review, 34*, 146–168.

Rindova, V. P., Williamson, I. O., Petkova, A. P., & Sever, J. M. (2005). Being good or being known: An empirical examination of the dimensions, antecedents, and consequences of organizational reputation. *Academy of Management Journal, 48*, 1033–1049.

Ritala, P. (2012). Coopetition strategy–when is it successful? Empirical evidence on innovation and market performance. *British Journal of Management, 23*, 307–324.

Ritala, P., & Sainio, L.-M. (2014). Coopetition for radical innovation: Technology, market and business-model perspectives. *Technology Analysis Strategic Management, 26*, 155–169.

Robert, M., Chiambaretto, P., Mira, B., & Le Roy, F. (2018). Better, faster, stronger, the impact of market oriented coopetition on product commercial performance. *Megye@N@ gement*, 574–610.

Rodrigues, M. B., Loureiro, S. M. C., & Romero, M. I. R. (2024). Luxury fashion games are so cool! predicting awareness, perceived quality, and loyalty. *Journal of Retailing and Consumer Services, 77*, Article 103668.

Rogers, P. (2023). 59+ Video Gaming Statistics For 2024 (Users & Popular Games). Answer IQ, answeriq.com.

Ruohonen, J., & Hyrynsalmi, S. (2017). Evaluating the use of Internet search volumes for time series modeling of sales in the video game industry. *Electronic Markets, 27*, 351–370.

Rusko, R. (2015). *Coopetition for organizations, encyclopedia of information science and technology, third edition* (pp. 576–586). IGI Global.

Sabri, O., Djedidi, A., & Hani, M. (2021). When does coopetition affect price unfairness perception? The roles of market structure and innovation. *Journal of Business Industrial Marketing, 36*, 209–229.

Sacranie, J. (2010) Consumer Perceptions & Video Game Sales: A Meeting of the Minds.

Shapiro, C. (1983). Premiums for high quality products as returns to reputations. *The Quarterly Journal of Economics, 98*, 659–679.

Statista. (2023). Video Games. Statista.

Statista. (2025). Number of games released on Steam worldwide from 2004 to 2024. Statista, statista.com.

Su, B.-C. (2007). Consumer e-tailer choice strategies at on-line shopping comparison sites. *International Journal of Electronic Commerce, 11*, 135–159.

Sun, X., Zhang, Y., & Feng, J. (2024). Impact of online information on the pricing and profits of firms with different levels of brand reputation. *Information Management, 61*, Article 103882.

Susanti, V., & Samudro, A. (2022). Assessing the role of corporate reputation on brand satisfaction: A study of chemical industry. *Corporate Reputation Review, 25*, 122–138.

Tang, Z., & He, B. (2022). The influence of information configuration on mobile game download. *Internet Research, 32*, 1191–1213.

Thombs, L. A., & Schucany, W. R. (1990). Bootstrap prediction intervals for autoregression. *Journal of the American Statistical Association, 85*, 486–492.

Tsiotsou, R. (2006). The role of perceived product quality and overall satisfaction on purchase intentions. *International Journal of Consumer Studies, 30*, 207–217.

Ubee, R. (2022). Foreign Censorship, Part 2: Trade and Economic Effects on U.S. Businesses, 2022 ed, United States International Trade Commission.

Un, C. A., & Asakawa, K. (2015). Types of R&D collaborations and process innovation: The benefit of collaborating upstream in the knowledge chain. *Journal of Product Innovation Management, 32*, 138–153.

Un, C. A., Cuervo-Cazurra, A., & Asakawa, K. (2010). R&D collaborations and product innovation. *Journal of Product Innovation Management, 27*, 673–689.

Van Beers, C., & Zand, F. (2014). R&D cooperation, partner diversity, and innovation performance: an empirical analysis. *Journal of Product Innovation Management, 31*, 292–312.

Van Der Merwe, A. W., & Puth, G. (2014). Towards a conceptual model of the relationship between corporate trust and corporate reputation. *Corporate Reputation Review, 17*, 138–156.

Veh, A., Göbel, M., & Vogel, R. (2019). Corporate reputation in management research: A review of the literature and assessment of the concept. *Business Research, 12*, 315–353.

Walker, K. (2010). A systematic review of the corporate reputation literature: Definition, measurement, and theory. *Corporate Reputation Review, 12*, 357–387.

Watson, F., & Wu, Y. (2022). The impact of online reviews on the information flows and outcomes of marketing systems. *Journal of Macromarketing, 42*, 146–164.

Wells, J. D., Valacich, J. S., & Hess, T. J. (2011). What signal are you sending? How website quality influences perceptions of product quality and purchase intentions. *MIS Quarterly*, 373–396.

Wesley II, C. L., Kong, D. T., Lubojacky, C. J., Saxton, M. K., & Saxton, T. (2022). Will the startup succeed in your eyes? Venture evaluation of resource providers during entrepreneurs' informational signaling. *Journal of Business Venturing, 37*, Article 106229.

Wiener, M., & Saunders, C. (2014). Forced coopetition in IT multi-sourcing. *The Journal of Strategic Information Systems, 23*, 210–225.

Xie, K. L., Xie, K. L., Zhang, Z., Zhang, Z., Zhang, Z., Zhang, Z., Singh, A., Singh, A., Lee, S. K., & Lee, S. K. (2016). Effects of managerial response on consumer eWOM and hotel performance: Evidence from TripAdvisor. *International Journal of Contemporary Hospitality Management, 28*, 2013–2034.

Yi, J., Lee, Y., & Kim, S.-H. (2019). Determinants of growth and decline in mobile game diffusion. *Journal of Business Research, 99*, 363–372.

Yuen, K. F., Thai, V. V., & Wong, Y. D. (2016). Are customers willing to pay for corporate social responsibility? A study of individual-specific mediators. *Total Quality Management Business Excellence, 27*, 912–926.

Zapadka, P., Hanelt, A., & Firk, S. (2022). Digital at the edge–antecedents and performance effects of boundary resource deployment. *The Journal of Strategic Information Systems, 31*, Article 101708.

Zhu, F., & Zhang, X. (2006). The influence of online consumer reviews on the demand for experience goods: the case of video games. *ICIS 2006 Proceedings, 25*.

Zhu, F., & Zhang, X. (2010). Impact of online consumer reviews on sales: The moderating role of product and consumer characteristics. *Journal of Marketing, 74*, 133–148.

Zulu-Chisanga, S., Nalmpanti, A. D., Boso, N., Hultman, M., & Leonidou, C. N. (2025). From rivals to allies: Building and benefiting from coopetition capability. *Industrial Marketing Management, 126*, 85–100.